



Group Members

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Guide Name

Dr. A. S. Mokhade, sir

Practical Title:

CSP510 Introduction to System Engineering:
Assignment 01: Discovering Systems Around Us

The Indian Railway

Part B1

Q1. What is the purpose of this system?

The purpose of the system: The Indian Railway is to provide transportation services to

Q2. Who are the stakeholders?

Q3. What are the major components?

Q4. What interactions exist?

Q5. What information flows?

Q6. What would happen if one component disappeared? Write for at least 3 such components.

Part B2

Q1. Do the parts affect each other?

Q2. Do the parts together produce an effect that is different from the effect of each part on its own?

Q3. Does the effect, the behavior over time, persist in a variety of circumstances?

Q4. If one component is improved significantly, will the entire system necessarily improve? Why?

Part C

Please classify into:

Physical:

Digital:

Human:

Social:

CyberPhysical:

System-of-Systems:

Adaptive Learning:

Part D
Boundary Identification for all the chosen systems

Part E

Q. Which property of this system cannot be explained by studying any single component individually?

Part F

Concept, Strategy, Operation analysis of these five systems:

Concept:

- **Why does this system exist?**

- **What human or societal problem does it solve? Strategy:**

- **Why is it organized this way?**

- **What alternative designs could have been used? Operations:**

- **How does the system function?**

- **How does it adapt?**

- **What failures may occur?**

Part G

What AI can be part of these systems to augment their functionality?

Part H

Reflection on how your definition and understanding of a system changed?

A Smartphone

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